

2D-DOA Estimation in Uniform Circular Arrays in the Presence of Mutual Coupling

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Abstract—This paper introduces an innovative iterative auto-calibration scheme for jointly estimating the two-dimensional (2-D) direction-of-arrival (DOA) and the mutual coupling coefficients for a signal measured using uniform circular arrays. The proposed method, first, leverages an integrated wideband dictionary to mitigate the detrimental effects of the discretization of the continuous parameter space over the considered azimuth and elevation angles. This approach enhances DOA estimation accuracy while significantly reducing computational complexity. With the improved reliability of DOA estimates, in the second step, the method also enables a more accurate estimation of mutual coupling coefficients. By iteratively alternating between these steps, the proposed scheme achieves high accuracy and robustness. In fact, the DOA estimation step utilizes an integrated dictionary in order to iteratively refine the active parameter space, thereby reducing the required computational complexity while maintaining overall performance. The complexity is further reduced by employing only the dominant subspace of the measured signal. Furthermore, the proposed method relies neither on prior knowledge of the number of nonzero coupling coefficients nor encounters ambiguity issues. Simulation results demonstrate the effectiveness of the proposed method.

Index Terms—Direction-of-Arrival, Sensor Array Processing, Sparse Representation, Integrated Dictionary, Auto-Calibration, Uniform Circular Array, Mutual Coupling Matrix.

I. INTRODUCTION

Estimating the direction-of-arrival (DOA) of a plane wave impinging on a sensor array has long been a cornerstone of array signal processing and remains a widely researched topic [1], [2]. In contrast to most studies, we address the practical scenario in which the array manifold is affected by mutual coupling, leading to a significant degradation in the performance of the applied algorithms [3].

The first approach to addressing mutual coupling involves offline calibration algorithms [4], [5], which rely on calibration sources with precisely known locations to determine the array manifold matrix. In contrast, the second approach employs auto-calibration or online calibration algorithms [6], [7], which eliminate the need for calibration sources by automatically compensating for the calibration. Since obtaining additional calibration sources is often impractical in many real-world applications, auto-calibration algorithms are generally preferred.

Additionally, the discretization of the parameter space introduces undesirable off-grid effects, which has recently increased interest in continuous parameter space techniques [8], [9]. An alternative approach involves utilizing a continuous dictionary in combination with an atomic norm penalty [10]–[12]. Although this approach often achieves accurate signal reconstruction, it typically results in optimization problems with high computational complexity. In this work, we propose leveraging an integrated wideband dictionary, initially introduced by Butsenko et al. [13] for frequency estimation, and designed over subsets of the continuous parameter space.

Common challenges with earlier auto-calibration methods include ambiguity problems [7], [14] and the requirement for a priori knowledge of the number of nonzero mutual coupling coefficients [15], both of which are widespread in practical applications. The proposed method, which iteratively estimates both the DOAs and the mutual coupling coefficients, effectively addresses these shortcomings. In the DOA estimation stage, a wideband dictionary framework is utilized in place of a conventional grid-based dictionary, which would otherwise demand high computational complexity to achieve a sufficiently fine grid. Next, the mutual coupling coefficients estimation is done utilizing the complex symmetric circular Toeplitz structure of the coefficients. These two steps are repeated iteratively until convergence or the desired accuracy is achieved. The matrix that represents the mutual coupling coefficients is referred to as the mutual coupling matrix (MCM).

The remainder of the paper is organized as follows. Section II outlines the data model for 2-D UCAs. Section III describes the details of the proposed method for joint iterative 2-D DOA and MCM estimation. Section IV provides some simulations to demonstrate the effectiveness of the proposed method. Finally, Section V concludes the work.

II. SIGNAL MODEL

Consider K narrowband far-field sources impinging simultaneously on a uniform circular array (UCA) of omnidirectional sensors. The angles of arrival are given by (ϕ, θ) , where ϕ represents the azimuth angle ($\phi \in [0, 360)$ degrees), measured counterclockwise from the x -axis, and θ represents the elevation angle ($\theta \in [0, 90]$ degrees), measured downward

from the z -axis. The UCA geometry, consisting of N equispaced sensors positioned along the circumference of the array, is depicted in Fig. 1 within the xy -plane. The marked arrow illustrates an example of the geometrical representation of a received signal.

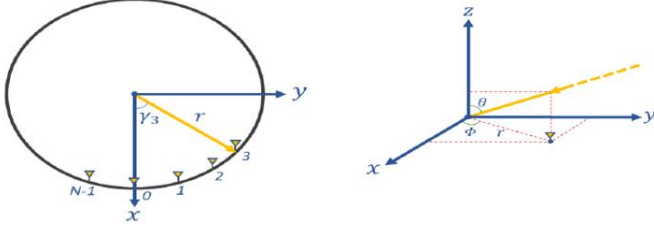


Fig. 1. Illustration of a UCA receiver model used for DOA and MCM estimation.

Considering an array of N sensors uniformly placed along the circumference of a circle with radius r , the array output $\mathbf{x} \in \mathbb{C}^{N \times T}$ can be expressed as

$$\mathbf{x}(t) = \mathbf{A}(\phi, \theta) \mathbf{s}(t) + \mathbf{e}(t), \quad t = t_1, t_2, \dots, t_T, \quad (1)$$

where $\mathbf{A}(\phi, \theta) = [\mathbf{a}(\phi_1, \theta_1), \mathbf{a}(\phi_2, \theta_2), \dots, \mathbf{a}(\phi_K, \theta_K)]$ represents the array manifold (steering matrix), with

$$\mathbf{a}(\phi_i, \theta_i) = \exp(j\xi_i \cos(\phi_i - \gamma_n)), \quad n = 0, 1, \dots, N-1,$$

denoting the steering vector corresponding to the source at (ϕ_i, θ_i) . Here, $\gamma_n = \frac{2\pi n}{N}$ represents the angular displacement of the n -th array element from the x -axis, and $\xi_i = k_0 r \sin \theta_i$, where $k_0 = \frac{2\pi}{\lambda}$ is the wave number. Additionally, $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_K(t)]^T$ and $\mathbf{e}(t) = [e_1(t), e_2(t), \dots, e_N(t)]^T$ represent the source signal vector and the additive white Gaussian noise vector, respectively. Finally, t denotes the snapshot index, and T represents the total number of snapshots. The signal vector $\mathbf{s}(t)$ and the noise vector $\mathbf{e}(t)$ are assumed to be zero-mean, circularly symmetric, and statistically independent random processes.

III. THE PROPOSED AUTOCALIBRATION METHOD

When mutual coupling is taken into account, the observation model in (1) can be reformulated as

$$\mathbf{X} = \mathbf{C} \mathbf{A}(\phi, \theta) \mathbf{S} + \mathbf{E}, \quad (2)$$

where $\mathbf{X} = [\mathbf{x}(t_1), \mathbf{x}(t_2), \dots, \mathbf{x}(t_T)]$, $\mathbf{S} = [\mathbf{s}(t_1), \mathbf{s}(t_2), \dots, \mathbf{s}(t_T)]$, and $\mathbf{E} = [\mathbf{e}(t_1), \mathbf{e}(t_2), \dots, \mathbf{e}(t_T)]$ represent the array output, the source signal matrix, and the noise data matrix, respectively. Additionally, $\mathbf{C}$ denotes the mutual coupling matrix, which is assumed to have a complex symmetric circular Toeplitz structure such that

$$\mathbf{C} = \text{toeplitz}(\tilde{\mathbf{c}}, \tilde{\mathbf{c}}) \in \mathbb{C}^{N, N}, \quad (3)$$

where

$$\tilde{\mathbf{c}} = [c_1, c_2, \dots, c_L, c_{L-1}, \dots, c_3, c_2] \in \mathbb{C}^{1, N}, \quad L = \frac{N}{2} + 1, \text{ for } N \text{ even,}$$

and

$$\tilde{\mathbf{c}} = [c_1, c_2, \dots, c_L, \dots, c_3, c_2] \in \mathbb{C}^{1, N}, \quad L = \left(\frac{N+1}{2}\right), \text{ for } N \text{ odd.}$$

Here, L denotes the degrees of freedom of the MCM, which corresponds to the number of unknown elements in the first row of the matrix $\mathbf{C}$. It is worth noting that, due to the inverse relationship between the magnitude/effect of mutual coupling and the distance between sensor pairs, the coefficients satisfy $0 < |c_L| < \dots < |c_2| < |c_1| = 1$.

In this section, we describe the overall structure of the proposed auto-calibration method for jointly estimating the 2-D DOAs and the mutual coupling matrix (MCM) of a UCA. For clarity, the method is organized into the following key components:

- **III-A: Sparse SVD-based Representation:** The measured data is first projected onto its dominant signal subspace using singular value decomposition (SVD). This step reduces the data dimension and computational complexity while preserving the essential signal information.
- **III-B: DOA Estimation:** An integrated wideband dictionary is constructed over the 2-D angular domain (azimuth and elevation). Initially, a coarse grid is used to mitigate off-grid effects. Then, a zooming procedure is applied iteratively to refine the regions containing active sources, leading to high-resolution DOA estimates.
- **III-C: MCM Estimation:** Exploiting the complex symmetric circular Toeplitz structure of the mutual coupling matrix, the MCM is estimated by minimizing a cost function that leverages the orthogonality between the noise subspace and the coupled array manifold.

These components are iterated until convergence or until the desired accuracy is reached, resulting in a robust and efficient joint estimation of both the DOAs and the MCM.

A. Sparse SVD-based Representation

The overall complexity of the proposed method can be significantly reduced by projecting the measured data onto the K -dimensional signal subspace, which is formed using the K dominant singular vectors of $\mathbf{X}$ [16]. The singular value decomposition (SVD) is performed as follows:

$$\mathbf{X} = \mathbf{U}_s \Sigma_s \mathbf{V}_s^H + \mathbf{U}_c \Sigma_c \mathbf{V}_c^H, \quad (4)$$

here, $\mathbf{U}_s \in \mathbb{C}^{N \times K}$ and $\mathbf{V}_s \in \mathbb{C}^{T \times K}$ represent the singular vectors corresponding to the K largest singular values, where K denotes the number of signals (sources). Define $\mathbf{X}_{SV} \triangleq \mathbf{X} \mathbf{V}_s$, $\mathbf{S}_{SV} \triangleq \mathbf{S} \mathbf{V}_s$, and $\mathbf{E}_{SV} \triangleq \mathbf{E} \mathbf{V}_s$, such that

$$\mathbf{X}_{SV} = \mathbf{C} \mathbf{A}(\phi, \theta) \mathbf{S}_{SV} + \mathbf{E}_{SV}, \quad (5)$$

By utilizing the $N \times K$ dimensional matrix $\mathbf{X}_{SV}$ instead of $\mathbf{X}$, the computational complexity can be significantly reduced. This reduction becomes particularly substantial when the number of sources is small and/or the number of time samples is large.

B. DOA Estimation

To determine the DOAs, let $(\phi_i, \theta_i)_{i=1}^Q$, for $i = 1, 2, \dots, Q$, represent the set of potential DOA candidates, where Q denotes the total number of candidates. Denote the dictionary over the set of candidates as $\mathbf{A}(\hat{\phi}, \hat{\theta}) = [\mathbf{a}(\hat{\phi}_1, \hat{\theta}_1), \mathbf{a}(\hat{\phi}_2, \hat{\theta}_2), \dots, \mathbf{a}(\hat{\phi}_Q, \hat{\theta}_Q)]$, where $\mathbf{a}(\hat{\phi}_i, \hat{\theta}_i) = [\mathbf{a}_0(\hat{\phi}_i, \hat{\theta}_i), \mathbf{a}_1(\hat{\phi}_i, \hat{\theta}_i), \dots, \mathbf{a}_{N-1}(\hat{\phi}_i, \hat{\theta}_i)]^T$. Each candidate is constructed using the integrated wideband element $\mathbf{a}_{n_r}(\phi, \theta)$, such that

$$\mathbf{a}_{n_r}(\phi_i, \theta_i) = \int_{\phi_i}^{\phi_{i+1}} \int_{\theta_i}^{\theta_{i+1}} \exp\left(jk_0 r \sin \theta \cos\left(\phi - \frac{2\pi n_r}{N}\right)\right) d\theta d\phi, \quad (6)$$

with $n_r = 0, 1, \dots, N-1$, where $[\phi_i, \phi_{i+1}]$ and $[\theta_i, \theta_{i+1}]$ represent the intervals for the i -th azimuth and elevation candidates, respectively. Indeed, instead of using a finely spaced narrowband dictionary of the candidate DOAs, the 2-D angular domain—comprising the azimuth $\phi : [0^\circ, 360^\circ)$ and elevation $\theta : [0^\circ, 90^\circ)$ —is divided into B_1 and B_2 continuous bands, respectively. As long as B_1 and B_2 are selected to be sufficiently large, this process would reduce the risk of any off-grid components and provide an initial coarse estimation of the angle of arrival. Since the dictionary encompasses all bands, no power remains off-grid, thereby avoiding a non-sparse solution caused by dictionary mismatch. In the initial step, this approach provides a coarse estimation of the regions of interest. Subsequently, the non-activated bands are discarded, and a refined dictionary is constructed over the activated bands. This process may then be repeated until one reaches the desired resolution. In essence, rather than constructing a dictionary over a fine grid—which may degrade performance due to off-grid effects and impose a significant computational burden—we employ the integrated dictionary to identify and iteratively refine the regions where sources are located.

Remark: In Appendix A, we introduce an approach to form the 2-D integral in equation 6.

The process of zooming, demonstrated in multiple steps (three steps in this case), is illustrated in Fig. 2 for a single angle θ . As shown in [13], the wideband dictionary significantly reduces the average computation time compared to its narrowband counterpart. Suppose the arrow in Fig. 2 indicates the location of one of the sources. The zooming process can then be described in three stages as follows:

- 1) Initially, form an integrated dictionary over the coarse grid (θ_0).
- 2) Next, identify the regions that may contain a source (for instance, the right side of θ_1 represents a possible region).
- 3) Finally, compose an integrated dictionary with finer grids over the potential regions identified in step 2 (illustrated as the θ_2 region in the third step).

Proceeding, the azimuth and elevation estimation are formulated using the LASSO method [17], which addresses the

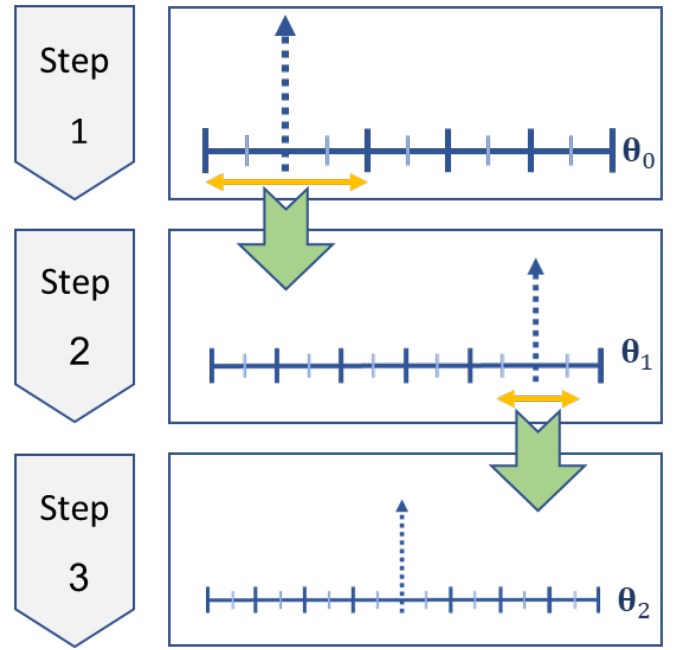


Fig. 2. Illustration of the zooming process over intervals/bands of angles.

resulting penalized regression problem. To this end, let $\hat{\mathbf{A}} = \hat{\mathbf{C}}\mathbf{A}(\hat{\phi}, \hat{\theta}) \in \mathbb{C}^{M \times Q}$, where $\mathbf{A}(\hat{\phi}, \hat{\theta})$ is the dictionary which shows the potential/possible DOAs. $\hat{\mathbf{C}}$ shows the estimated MCM. then

$$\mathbf{D} = \begin{bmatrix} \hat{\mathbf{A}} \\ \vdots \\ \hat{\mathbf{A}} \end{bmatrix}_{NK \times Q} \quad (7)$$

Then,

$$\begin{bmatrix} \mathbf{X}_{sv}(:, 1) \\ \vdots \\ \mathbf{X}_{sv}(:, K) \end{bmatrix} = \mathbf{D}\hat{\mathbf{s}} + \begin{bmatrix} \mathbf{N}_{sv}(:, 1) \\ \vdots \\ \mathbf{N}_{sv}(:, K) \end{bmatrix} \quad (8)$$

Let $\mathbf{x} = \text{vec}(\mathbf{X}_{sv}) \in \mathbb{C}^{NK}$ and $\mathbf{n} = \text{vec}(\mathbf{N}_{sv}) \in \mathbb{C}^{NK}$. Then,

$$\mathbf{x} = \mathbf{D}_{NK \times Q}\hat{\mathbf{s}} + \mathbf{n}, \quad (9)$$

this allows the LASSO formulation, where $\mathbf{s}_d$, which is the desired signal, is obtained as

$$\mathbf{s}_d = \arg \min_{\hat{\mathbf{s}}} \left\{ \|\mathbf{x} - \mathbf{D}\hat{\mathbf{s}}\|_2^2 + \gamma \|\hat{\mathbf{s}}\|_1 \right\}, \quad (10)$$

where $\hat{\mathbf{s}} \in \mathbb{C}^{Q \times 1}$, and γ is a user-defined parameter that controls the sparsity of the solution. In this context, γ is chosen as

$$\gamma = \alpha \max_i |\mathbf{a}_i^H \mathbf{x}|, \quad (11)$$

where $\mathbf{a}_i$, $i = 1, 2, \dots, Q$ are columns of the $\mathbf{D}$. The constant α is a user-defined parameter that enables evaluation of the performance of each dictionary (as shown in the simulation

results) by varying its value within the range $0 < \alpha \leq 1$. The parameter $\gamma_{\max} = \alpha \max_i |\mathbf{a}_i^H \mathbf{x}|$ represents the smallest tuning parameter value that results in a solution with all coefficients equal to zero [13], [17].

C. MCM Estimation

Proceeding, the mutual coupling matrix coefficients are estimated by exploiting the complex symmetric circular Toeplitz structure of the MCM in a UCA. Let $\mathbf{R}_x$ denote the array output covariance matrix, and let λ_n and $\mathbf{e}_n$ be its eigenvalues and corresponding eigenvectors, respectively. Then,

$$\mathbf{R}_x = \sum_{n=1}^N \lambda_n \mathbf{e}_n \mathbf{e}_n^H = \mathbf{E}_s \Lambda_s \mathbf{E}_s^H + \mathbf{E}_n \Lambda_n \mathbf{E}_n^H, \quad (12)$$

where $\mathbf{E}_s = [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_K] \in \mathbb{C}^{N \times K}$ represents the signal subspace containing the principal eigenvectors corresponding to the K largest eigenvalues. The noise subspace, $\mathbf{E}_n$, consists of the remaining $N - K$ eigenvectors.

The signal subspace spans the same space as the array manifold matrix, i.e., $\text{span}\{\mathbf{E}_s\} \perp \text{span}\{\mathbf{C}\mathbf{A}(\phi, \theta)\}$. Moreover, the signal subspace and noise subspace are orthogonal, i.e., $\text{span}\{\mathbf{E}_s\} \perp \text{span}\{\mathbf{E}_n\}$. This orthogonality implies that $\text{span}\{\mathbf{E}_n\} \perp \text{span}\{\mathbf{C}\mathbf{A}(\phi, \theta)\}$.

Assuming that the DOAs have been obtained (as described in Subsection III-B), then, the MCM can be estimated as the matrix $\mathbf{C}$ by minimizing:

$$J = \|\mathbf{E}_n^H \mathbf{C} \mathbf{A}(\phi, \theta)\|_F^2 = \sum_{k=1}^K \|\mathbf{E}_n^H \mathbf{C} \mathbf{a}(\phi_k, \theta_k)\|^2 \quad (13)$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

Leveraging the circular symmetry of the mutual coupling matrix, the estimation can instead focus on the vector that constructs the matrix. Specifically, this involves rewriting $\mathbf{C}\mathbf{a}(\phi, \theta) = \mathbf{F}\{\mathbf{a}(\phi, \theta)\}\mathbf{c} \in \mathbb{C}^{L \times 1}$, where $\mathbf{F}$ represents the resulting transform matrix, and $\mathbf{c} = [c_1, c_2, \dots, c_L]^T \in \mathbb{C}^L$ is the mutual coupling vector. By extending the relation of the mutual coupling introduced in [18], [19] to 2-D and exploiting the symmetric circulant property of $\mathbf{C}$, the matrix $\mathbf{F}\{\mathbf{a}(\phi, \theta)\}$ is defined as:

$$\mathbf{F}\{\mathbf{a}(\phi, \theta)\} = \mathbf{F}_1\{\mathbf{a}(\phi, \theta)\} + \mathbf{F}_2\{\mathbf{a}(\phi, \theta)\} + \mathbf{F}_3\{\mathbf{a}(\phi, \theta)\} + \mathbf{F}_4\{\mathbf{a}(\phi, \theta)\} \quad (14)$$

$$\begin{aligned} [\mathbf{F}_1\{\mathbf{a}(\phi, \theta)\}]_{i,j} &= \begin{cases} [\mathbf{a}(\phi, \theta)]_{i+j-1,1} & \text{if } i+j \leq N+1, \\ 0 & \text{otherwise,} \end{cases} \\ [\mathbf{F}_2\{\mathbf{a}(\phi, \theta)\}]_{i,j} &= \begin{cases} [\mathbf{a}(\phi, \theta)]_{j-i+1,1} & \text{if } i \geq 2, \\ 0 & \text{otherwise,} \end{cases} \\ [\mathbf{F}_3\{\mathbf{a}(\phi, \theta)\}]_{i,j} &= \begin{cases} [\mathbf{a}(\phi, \theta)]_{N+i+j-1,1} & \text{if } i < j \leq p, \\ 0 & \text{otherwise,} \end{cases} \\ [\mathbf{F}_4\{\mathbf{a}(\phi, \theta)\}]_{i,j} &= \begin{cases} [\mathbf{a}(\phi, \theta)]_{i+j-N-1,1} & \text{if } 2 \leq j \leq p, i+j \geq N+2, \\ 0 & \text{otherwise.} \end{cases} \end{aligned} \quad (15)$$

where $i = 1, 2, \dots, N$, $j = 1, 2, \dots, L$, and $p = \lfloor \frac{N+1}{2} \rfloor$. Applying this transformation to (13) yields

$$J = \sum_{k=1}^K \mathbf{c}^H \mathbf{F}^H \{\mathbf{a}(\phi_k, \theta_k)\} \mathbf{E}_n \mathbf{E}_n^H \mathbf{a}(\phi_k, \theta_k) \mathbf{F} \{\mathbf{a}(\phi_k, \theta_k)\} \mathbf{c} = \mathbf{c}^H \mathbf{U}(\phi, \theta) \mathbf{c}, \quad (16)$$

where

$$\mathbf{U}(\phi, \theta) = \sum_{k=1}^K \mathbf{F}^H \{\mathbf{a}(\phi_k, \theta_k)\} \mathbf{E}_n \mathbf{E}_n^H \mathbf{F} \{\mathbf{a}(\phi_k, \theta_k)\}^H \in \mathbb{C}^{L \times L}, \quad (17)$$

By imposing the constraint that $c_1 = 1$, (16) can be utilized as the cost function to be minimized in order to estimate the desired mutual coupling vector, i.e.,

$$\mathbf{c} = \mathbf{e}_{\min} [\mathbf{U}(\phi, \theta)], \quad (18)$$

where $\mathbf{e}_{\min}[\cdot]$ denotes the eigenvector associated with the smallest eigenvalue. These two steps in sections III-B and III-C are iteratively performed until the process converges or reaches the targeted accuracy. It is worth noting that the DOA estimation process itself is also iterative.

In summary, Algorithm 1 outlines the proposed method for jointly estimating the DOA and the mutual coupling matrix (MCM).

Algorithm 1 The proposed SVD-based sparse WB method

- 1: Assign an initial MCM such that $\mathbf{C}_0 = \mathbf{I}_{N \times N}$.
 - 2: Choose the number of zooming steps and number of bands to use for both azimuth and elevation; select a suitable α for Eq. (11).
 - 3: Take the SVD of Eq. (2).
 - 4: Compose the DOA integrated wideband dictionary with elements defined in Eq. (6).
 - 5: Estimate the azimuth and elevation (as stated in Subsection III-B) using Eq. (10).
 - 6: Estimate the MCM as detailed in Subsection III-C.
 - 7: Form the DOA estimates on the active bands (composing of the zoomed dictionary given in Eq. (6)), and then using Eq. (10) by considering the MCM in step 6. The iterative zooming procedure enables discarding the non-activated parts of each dictionary.
 - 8: Estimate the MCM as stated in Subsection III-C.
 - 9: Repeat steps 7 and 8 until obtaining the desired resolution.
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IV. SIMULATION RESULTS

In this section, simulations are conducted to evaluate the performance of the proposed method. We consider a UCA consisting of $N = 15$ sensors, with a radius of $r = \lambda$, where λ represents the center wavelength of the narrowband signals. We further assume three far-field sources with angles of arrival $(18.3^\circ, 243.4^\circ)$, $(83.6^\circ, 60^\circ)$, and $(73.9^\circ, 357.8^\circ)$, for which $T = 200$ snapshots are measured. The signals and the additive sensor noises are stationary, mutually uncorrelated, and have zero mean. The signals are modeled as circularly symmetric Gaussian processes with a variance of σ_i^2 for the i -th process. The corrupting noise is assumed to be i.i.d. white circularly symmetric Gaussian process with variance σ^2 , such that the input SNR for the i -th signal is given by $10 \log(\sigma_i^2/\sigma^2)$. The coupling coefficients are set to $c_2 = 0.79 + j0.432$ and $c_3 = 0.35 + j0.16$.

Fig. 3 presents the probability of correctly estimating the number of angle of arrivals using an integrated wideband dictionary. Let B_i^θ and B_i^ϕ represent the number of bands for elevation and azimuth, respectively, where $i = 1, 2, 3$ corresponds to each stage of the dictionary. For the zooming steps in elevation, we consider $B_1^\theta = 30$, $B_2^\theta = 10$, $B_3^\theta = 3$, and for azimuth $B_1^\phi = 120$, $B_2^\phi = 10$, and $B_3^\phi = 3$ (depicted by the grey curve). Additionally, another configuration is considered with $B_1^\theta = 30$, $B_2^\theta = 5$, $B_3^\theta = 3$, $B_1^\phi = 120$, $B_2^\phi = 5$, and $B_3^\phi = 3$ (depicted by the brown curve). This figure examines the correct model order for varying values of α as defined in (11). As evident from Fig. 3, using $\alpha \leq 0.4$ for the three-stage dictionary (grey curve) represents a suitable region for applying the proposed method.

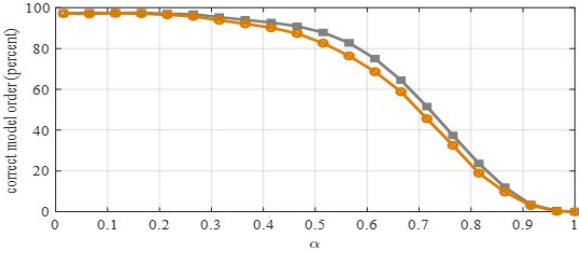


Fig. 3. Choosing suitable α for equation (11).

To evaluate the accuracy of DOA estimation methods, the average root-mean-square errors (RMSEs) for the elevation and azimuth estimates is calculated as follows:

$$\text{RMSE} = \sqrt{\frac{1}{KM} \sum_{k=1}^K \sum_{m=1}^M [(\theta_k^m - \theta_k)^2 + (\phi_k^m - \phi_k)^2]}, \quad (19)$$

where $M = 500$ represents the number of Monte Carlo simulations, and (θ_k^m, ϕ_k^m) denotes the estimate of (θ_k, ϕ_k) in the m -th Monte Carlo run (out of M runs). Fig. 4 illustrates the resulting RMSE as the SNR varies from 0 dB to 20 dB. The results are compared with LASSO estimates obtained using a dictionary composed of discrete prolate spheroidal sequences (DPSS) [20], [21], as well as with the 2-D extension of the method proposed in [13], referred to as Iterative-MUSIC method. This Iterative-MUSIC is found to have the highest computational complexity while delivering the poorest performance of the discussed methods.

Fig. 5 illustrates the RMSE of the coupling coefficients, defined as

$$\text{RMSE}_{\text{coupling coeff.}} = \frac{\sqrt{\sum_{m=1}^M \|\hat{\mathbf{c}}^m - \mathbf{c}\|^2}}{\|\mathbf{c}\|} \times 100\%, \quad (20)$$

where $\hat{\mathbf{c}}^m$ represents the estimated coupling coefficient vector, $\mathbf{c}$, in the m -th Monte Carlo experiment. As shown in Fig. 5, the proposed method demonstrates high effectiveness and precision in estimating the coupling coefficients. As demonstrated

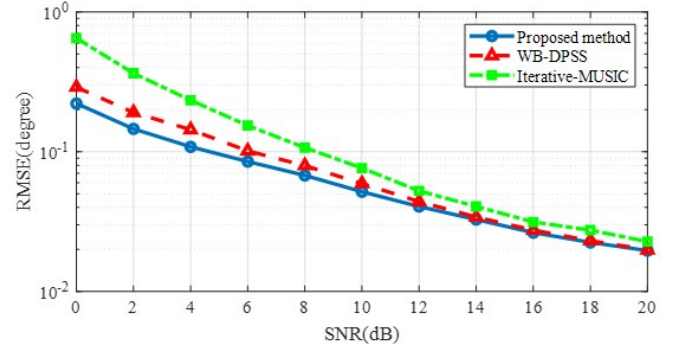


Fig. 4. The RMSE of angle estimation as a function of the SNR.

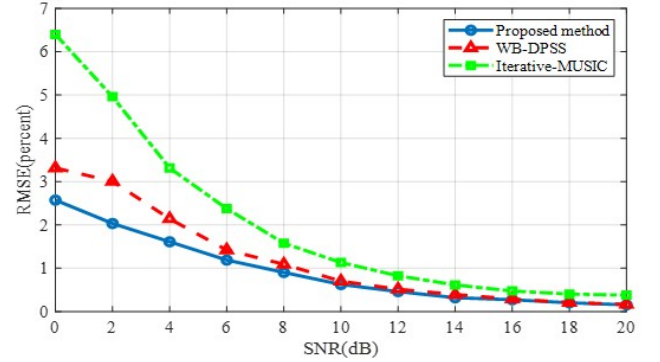


Fig. 5. The RMSE of mutual coupling coefficients estimation as a function of the SNR.

in the simulation results¹, the proposed method outperforms the other two methods.

V. CONCLUSION

In this work, we introduced a self-calibration algorithm capable of jointly estimating 2-D DOAs and the MCM for a UCA configuration. The DOA estimation step leverages a sparse representation framework combined with integrated dictionary elements that span bands of the desired parameter space. The MCM estimation step exploits the circular symmetry of the mutual coupling matrix specific to UCAs. This iterative procedure continues until the results converge or meet the desired level of accuracy.

The proposed method offers several advantages, including enabling initial coarse gridding of the dictionary without the risk of off-grid effects, imposing no constraints on the coupling matrix, and avoiding ambiguity in the resulting DOA estimates. Future work could explore extending the proposed algorithm to handle scenarios with nonuniform circular arrays or arrays with arbitrary geometries

¹Additional results are provided in the complete version at <https://arxiv.org/abs/2404.17089>.

APPENDIX A

SIMPLIFYING THE 2-D INTEGRAL OF THE DICTIONARY ELEMENTS IN ORDER TO DO SIMPLER NUMERICAL COMPUTATIONS

To simplify the 2-D integration in (6) and attain a formula for computing the integration, one may use the Taylor series expansion for e^z about $z = 0$, i.e.,

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad (21)$$

Let $\tau = k_0 r$, then

$$\begin{aligned} \nu &\triangleq \int_{\phi_i}^{\phi_{i+1}} \int_{\theta_i}^{\theta_{i+1}} e^{j\alpha \sin \theta \cos(\phi - \lambda_n)} d\theta d\phi \\ &= \int_{\phi_i}^{\phi_{i+1}} \int_{\theta_i}^{\theta_{i+1}} \sum_{n=0}^{\infty} \frac{(\tau \cos(\phi - \lambda_n))^n}{n!} \sin^n \theta d\theta d\phi \\ &= \int_{\phi_i}^{\phi_{i+1}} \sum_{n=0}^{\infty} \left(\frac{\tau^n}{n!} \cos^n(\phi - \lambda_n) \int_{\theta_i}^{\theta_{i+1}} \sin^n \theta d\theta \right) d\phi \\ &= \sum_{n=0}^{\infty} \frac{\tau^n}{n!} \int_{\phi_i}^{\phi_{i+1}} \cos^n(\phi - \lambda_n) d\phi \int_{\theta_i}^{\theta_{i+1}} \sin^n \theta d\theta \\ &= \sum_{n=0}^{\infty} \frac{\tau^n}{n!} \int_{\phi_i - \lambda_n}^{\phi_{i+1} - \lambda_n} \cos^n \phi d\phi \int_{\theta_i}^{\theta_{i+1}} \sin^n \theta d\theta \quad (22) \end{aligned}$$

Expanding the first and the second terms yields

$$\begin{aligned} \nu &= \left(\int_{\phi_i - \lambda_n}^{\phi_{i+1} - \lambda_n} d\phi \right) \left(\int_{\theta_i}^{\theta_{i+1}} d\theta \right) + \tau \left(\int_{\phi_i - \lambda_n}^{\phi_{i+1} - \lambda_n} \cos \phi d\phi \right) \\ &\quad \left(\int_{\theta_i}^{\theta_{i+1}} \sin \theta d\theta \right) + \sum_{n=2}^{\infty} \frac{\tau^n}{n!} \int_{\phi_i - \lambda_n}^{\phi_{i+1} - \lambda_n} \cos^n \phi d\phi \\ &\quad \int_{\theta_i}^{\theta_{i+1}} \sin^n \theta d\theta \quad (23) \end{aligned}$$

The results of first and second terms are simple closed forms. Integration by parts for the third term yields

$$\int \sin^n \theta d\theta = -\frac{1}{n} \sin^{n-1} \theta \cos \theta + \frac{n-1}{n} \int \sin^{n-2} \theta d\theta \quad (24)$$

and

$$\int \cos^n \phi d\phi = \frac{1}{n} \cos^{n-1} \phi \sin \phi + \frac{n-1}{n} \int \cos^{n-2} \phi d\phi \quad (25)$$

One may thus avoid a numerical integration of that complex integrand by computing the recursive factors in the terms. To do this, the number of terms (for this work, computing about ~ 10 terms suffices and gives the same precision as the built-in `integral2` function in Matlab) in this series could be computed to get the required precision.

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